Dermatopathology Practice Questions and Answers

1. A 70-year-old male patient of Asian ethnicity presents with shortness of breath and imaging findings concerning for widespread metastatic lesions in the lungs and multiple lymph nodes. As there is no definitive primary source of malignancy identified by imaging, you realize you have not completed a thorough skin exam. What is the most likely source of the patient's metastatic disease that you expect to find on skin examination?

   A. Mycosis fungoides causing erythroderma
   B. Acral melanoma involving the plantar surface
   C. Basal cell carcinoma on the upper lip
   D. Solitary seborrheic keratosis on the trunk

The correct answer is "acral melanoma involving the plantar surface." While acral melanomas represent less than 5% of all melanomas, they are the most common type of melanoma in darker-pigmented patients. A disproportionate percentage of acral melanomas are diagnosed at advanced stage.

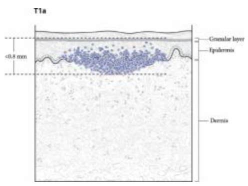
Mycosis fungoides is characterized by an indolent clinical course and excellent prognosis in limited stage disease. Basal cell carcinoma rarely metastasizes; a basal cell carcinoma occurring on the upper lip would presumably be more conspicuous and prompt management in contrast to an acral melanoma on the plantar surface. A solitary seborrheic keratosis is benign and would not produce widely metastatic lesions.

2. Which feature of melanoma on pathological examination correlates with worse clinical behavior?

   A. Amount of pigment within individual malignant melanocytes
   B. Size of tumor nests in the dermis
   C. Tumor thickness
   D. Presence of epidermal spongiosis
   E. Asymmetry

The correct answer is "tumor thickness." In melanoma, the risk of metastasis correlates with the tumor thickness.

(Extra info: tumor thickness is measured from the top of the granular layer of the epidermis to the deepest invasive cell at the base of the tumor:

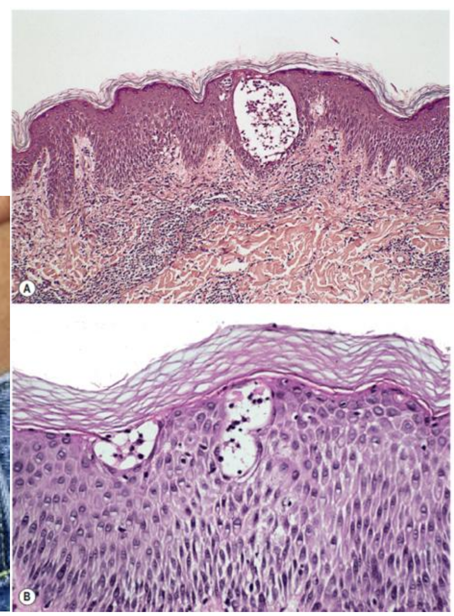


AJCC Cancer Staging System, version 9)

3. Regarding squamous cell carcinoma of the lip, where does it commonly occur, and what is the proposed etiology?
   A. Upper lip; due to BRAF mutation
   B. Lower lip; due to BRAF mutation
   C. Upper lip; due to arsenic exposure
   D. Lower lip; due to UV radiation

The correct answer is “Lower lip; due to UV radiation.” Cutaneous squamous cell carcinoma of the lip primarily occurs on the lower lip. The primary risk factor for cutaneous squamous cell carcinoma is UV light exposure.

4. What is the type of hypersensitivity reaction associated with the clinical and microscopic photographs below?



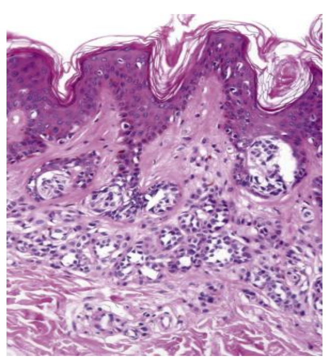
   A. Type I (atopic)
   B. Type II (antibody-mediated)
   C. Type III (immune complex)

D. Type IV (delayed)

The correct answer is “type IV (delayed)” hypersensitivity reaction. This type of response does not involve antibodies since it is cell mediated (as opposed to types I, II, and III which are antibody mediated). The images show allergic contact dermatitis that occurs after sensitization to certain metals, for example, such as the button on this patient’s jeans. Other examples of delayed hypersensitivity reactions include the tuberculin skin test (which uses purified-protein derivative or “PPD”).

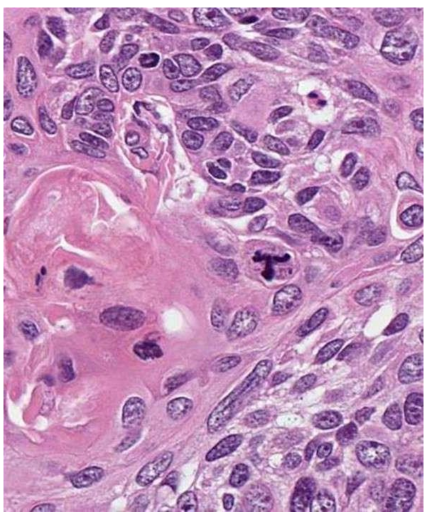
5. You identify a pigmented macule on the back of a 35-year-old female, as shown in the clinical image below, and decide to excise it as it measures just over 6 mm and exhibits some variegation in color. A representative H&E-stained section is shown in the microscopic image. Which of the following is true regarding the lesion?
   A. The lesion is keratinocytic
   B. These lesions only occur on sun-exposed sites
   C. The cells comprising the lesion have activating mutations in BRAF, but will undergo a permanent growth arrest in a great majority of cases
   D. The cells comprising the lesion have activating mutations in BRAF, and will progress to metastatic malignancy in most cases
   E. This lesion is a nodular melanoma

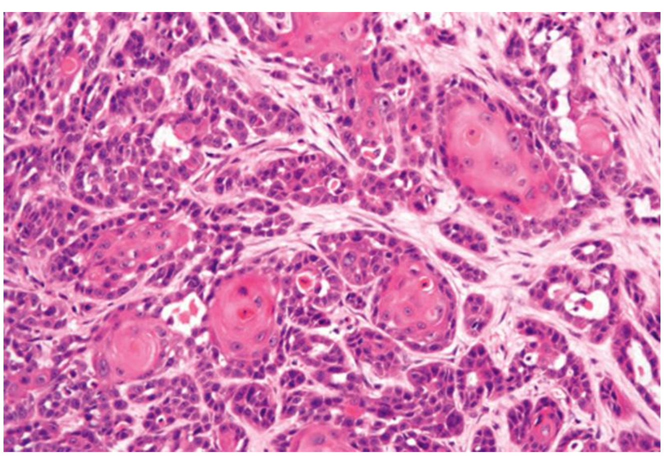
The correct answer is “the cells comprising the lesion have activating mutations in BRAF but will undergo a permanent growth arrest in a great majority of cases.” The images show a compound melanocytic nevus, which usually contain BRAF V600E activating mutations. However, in a majority of cases, there is growth arrest and no progression to melanoma. In the microscopic image, the compound melanocytic nevus is comprised of bland melanocytes forming nests at the tips of the rete ridges; reassuringly, there is maturation (or decrease in size) of melanocytes comprising the dermal part of lesion. Melanocytic nevi are concentrated in sun-exposed areas, but also occur in other sites (i.e., acral sites like the palms, soles, nail matrix).

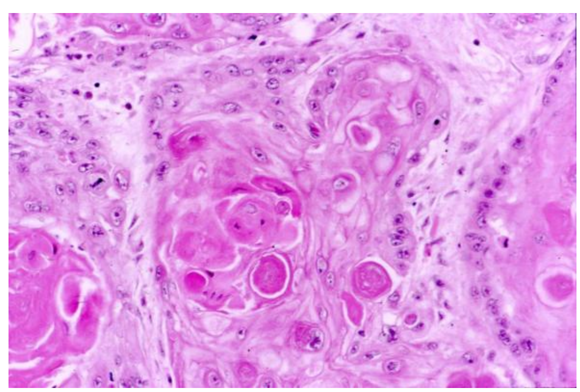


6. An 85-year-old man has a solitary raised, ulcerated nodule on his ear. The deep dermis is shown in the corresponding microscopic images. What is the most likely diagnosis?
   A. Melanoma in a radial growth phase
   B. Morphea-type basal cell carcinoma with a moderately high risk of metastasis
   C. Actinic keratosis
   D. Invasive squamous cell carcinoma with a low risk of metastasis
   E. Keratoacanthoma with a moderately high risk of metastasis

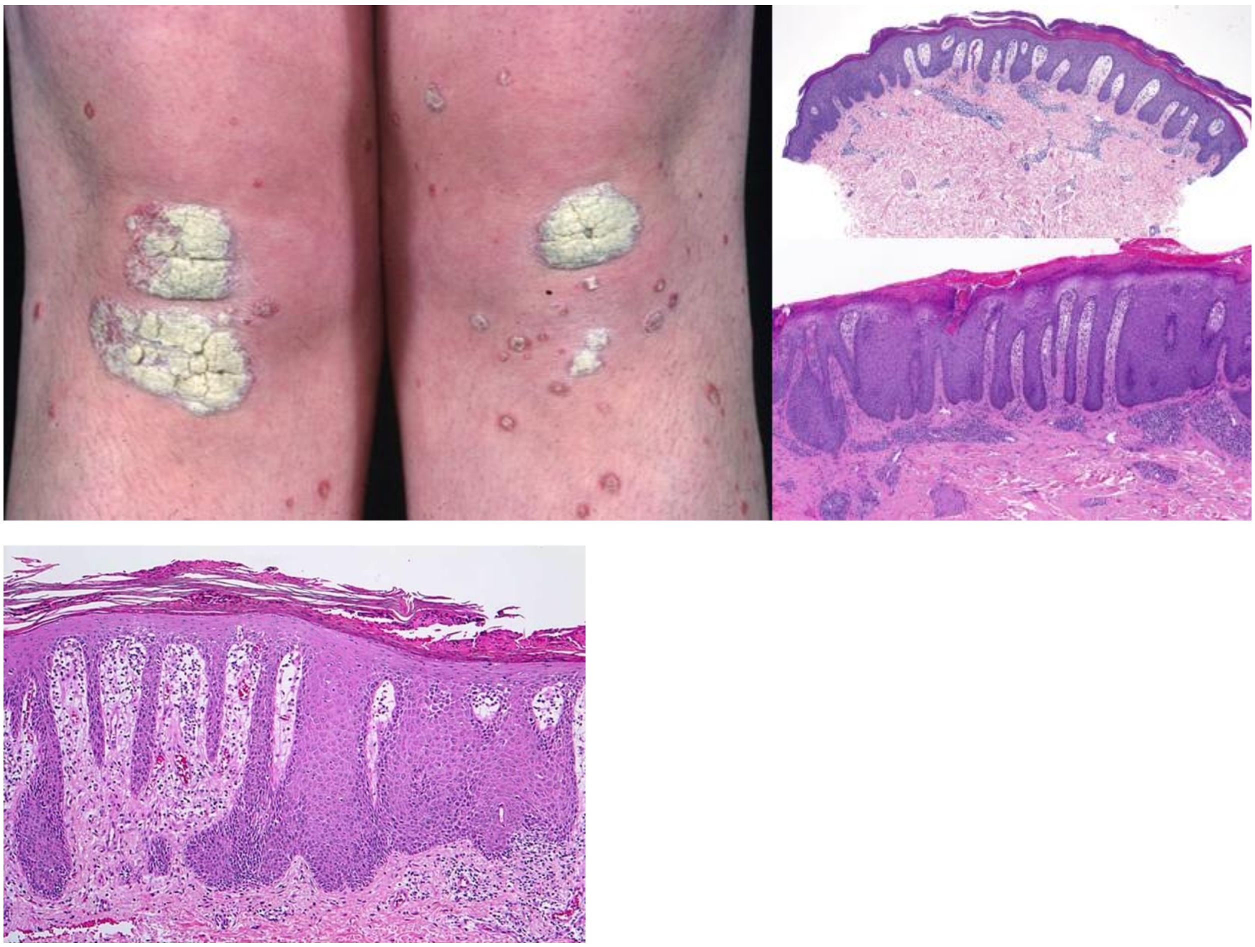
The correct answer is “invasive squamous cell carcinoma with a low risk of metastasis.” The images exhibit malignant squamous cells showing aberrant deep keratinization and keratin pearl formation. There is also an atypical tri-polar mitotic figure seen in the first image which additionally supports the diagnosis of malignancy and invasive squamous cell carcinoma. Approximately 2-5% of invasive squamous cell carcinomas metastasize.







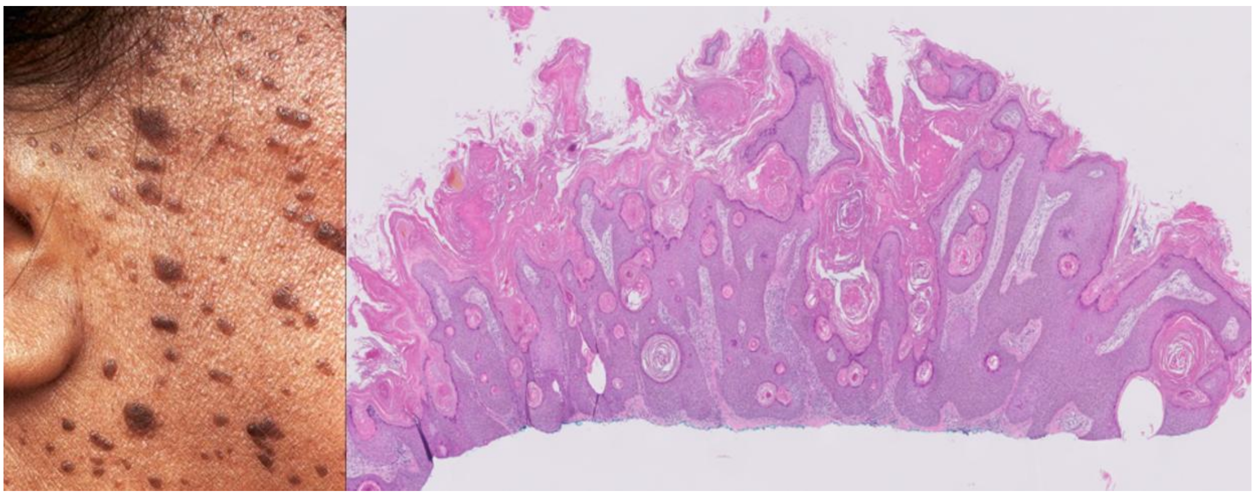
7. A 60-year-old woman has a long-standing history of erythematous, scaly, thickened patches on her knees, elbows, and back. She recalls developing arthritis involving her DIP joints after the onset of her cutaneous lesions. On physical exam, the cutaneous lesions appear to have a thick, silvery-white scale, and some of them exhibit pinpoint bleeding after removal of the scale. An image taken from her physical exam and corresponding microscopic images of a biopsy are shown below. What is the diagnosis?



A. Psoriasis
B. Lichen planus
C. Squamous cell carcinoma
D. Fixed drug eruption
E. Irritated seborrheic keratosis

The correct answer is “psoriasis.” The clinical history of the “erythematosquamous” cutaneous lesions with silvery scale followed by arthritis is compatible with psoriasis. Psoriatic arthritis may occur in up to 30% of patients with psoriasis. The microscopic images show regular psoriasiform epidermal hyperplasia (“test tubes in a rack”) with overlying parakeratotic cells (resulting from increased transit time), and dilation and increase in vessels in dermal papillae which are likely responsible for the tendency towards pinpoint bleeding when overlying scale is removed ("Auspitz sign").

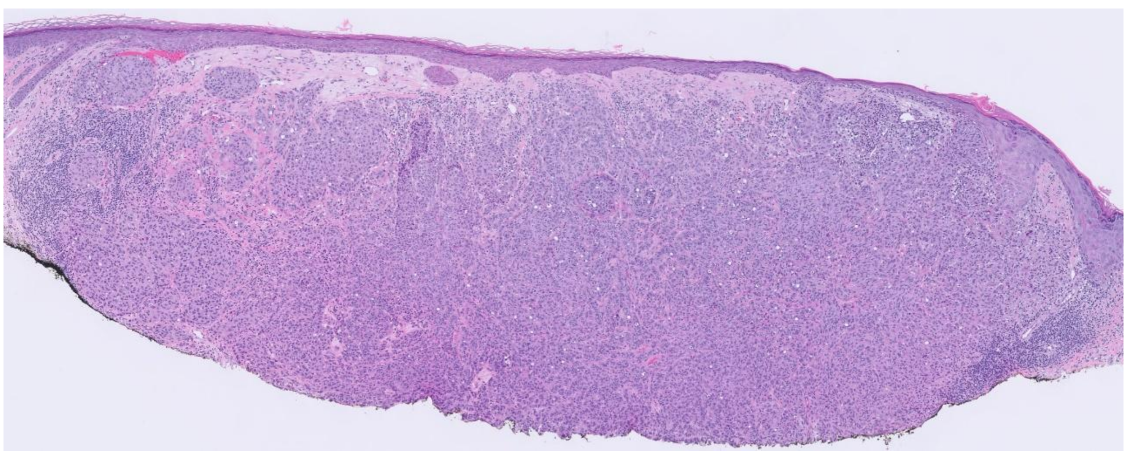
8. An 80-year-old woman presents with abdominal pain and weight loss. Pertinent features from the H&P include the development of numerous keratotic, exophytic papules and plaques that appear to be "stuck-on," and developed in the last 4 months (see clinical image and corresponding microscopic image of a biopsy). Which of the following is the most likely cause of her abdominal pain and weight loss?



A. Metastatic melanoma
B. Helicobacter pylori infection
C. Gastric adenocarcinoma
D. Basal cell carcinoma
E. Sézary syndrome

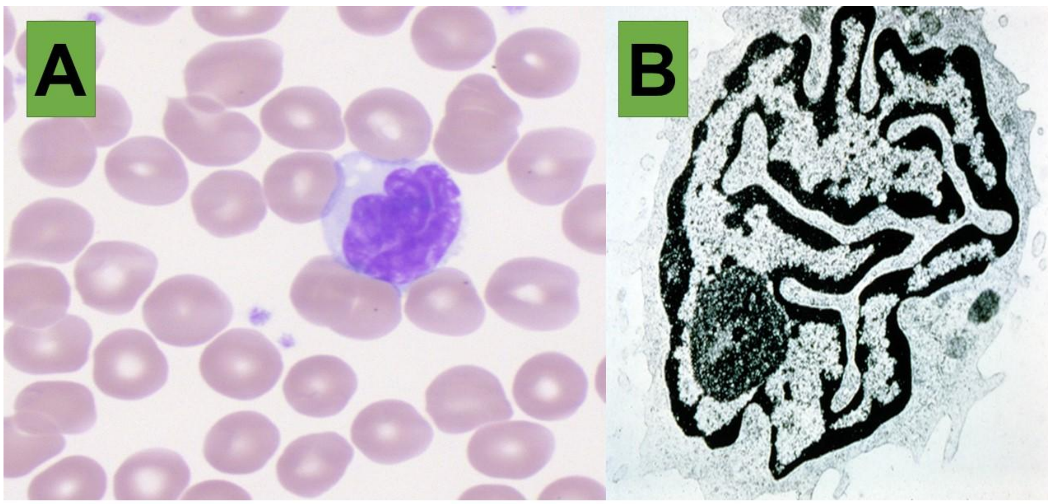
The correct answer is "gastric adenocarcinoma." The clinical scenario described is that of the sign of Leser-Trelat, which is the sudden onset of seborrheic keratoses as a sign of internal malignancy. The most common associated malignancies include gastrointestinal adenocarcinomas such as gastric adenocarcinoma, which is the best answer. While some of the lesions in the clinical image appear pigmented, the keratotic, "stuck-on" description argues against melanoma, as does the rapidity in onset.

9. What is the most likely diagnosis of the nodule in the clinical image below, measuring 1.2 cm, which shows multiple atypical mitotic figures on the corresponding microscopic image and exhibits only a vertical growth phase?
   A. Dermatofibroma
   B. Nodular melanoma
   C. Macular seborrheic keratosis
   D. Pustular psoriasis
   E. Verruca vulgaris



The correct answer is “nodular melanoma.” The size and clinical description are those of a nodule (given that the diameter is > 1 cm). Nodular melanoma will exhibit malignant growth, usually with multiple atypical mitotic figures, and only demonstrates a vertical growth phase. Superficial spreading melanoma and lentigo maligna melanoma show radial growth phases before entering a vertical growth phase.

10. A 64-year-old male presents with erythroderma. Physical exam is significant for generalized lymphadenopathy. His peripheral blood (PB) leukocyte count was 18,500 mm3. A differential count performed on a peripheral blood smear revealed circulating cells with morphologic features shown in image A. A transmission electron microscopic study of PB revealed cells shown in image B. Molecular studies reveal that these circulating cells are clonal. What are these cells?



A. Langerhans cells

B. Macrophages

C. Mast cells

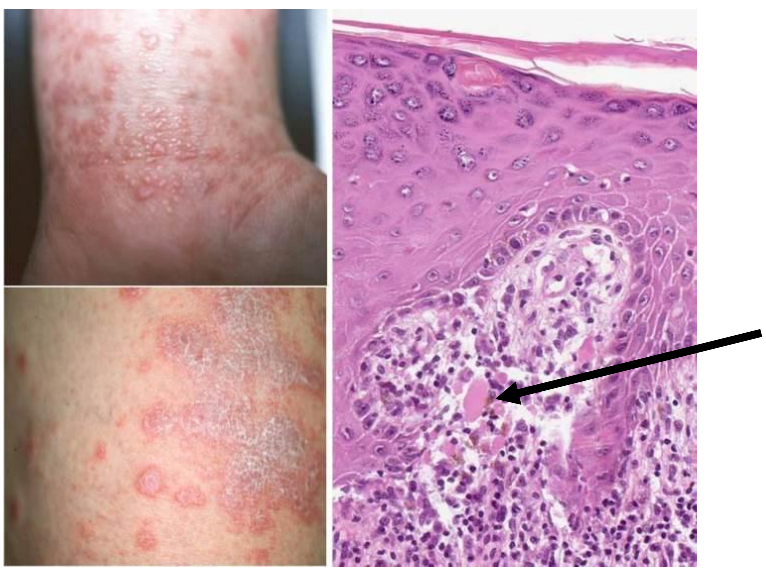
D. Neutrophils

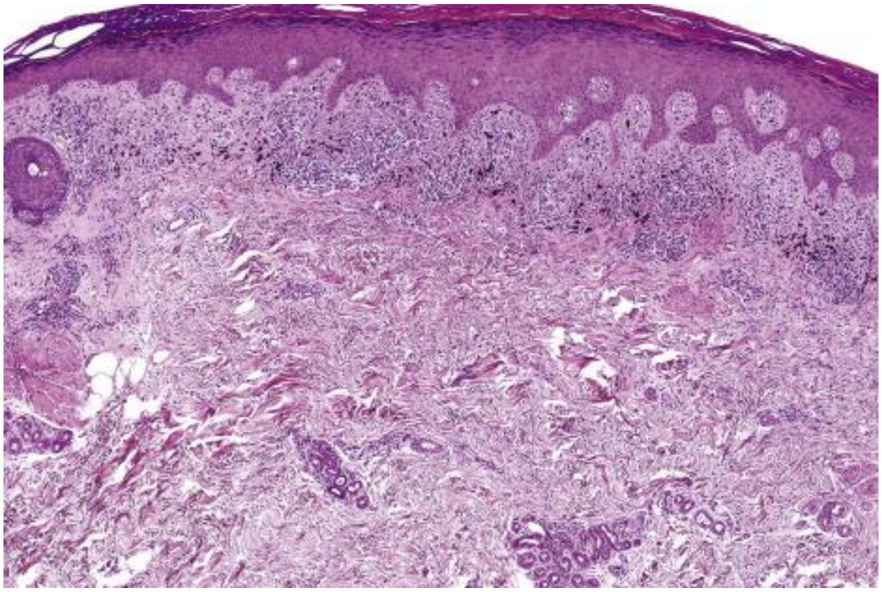
E. Sézary cells

The correct answer is “Sézary cells.” This is a syndrome defined by the presence of erythroderma, generalized lymphadenopathy, and atypical circulating lymphoid cells with cerebriform morphology that are clonal. The electron micrograph image shows a Sézary cell with invaginated nuclear membranes and nuclear clefting. The cell with the abnormal cerebriform nucleus in the EM image is a T-cell. Langerhans cells reside in the epidermis and would show tennis-racket-shaped Birbeck granules on electron microscopy. Macrophages, mast cells, and neutrophils are of the myeloid lineage and may show granules in the cytoplasm and are not the neoplastic cells responsible for the clinical scenario of Sézary syndrome.

11. A 45-year-old developed an intensely pruritic, polygonal, papular eruption with violaceous color on the flexor surfaces of his upper extremities. The papules transformed into plaques covered by fine white reticular scales. One of the lesions was biopsied and showed a band-like inflammatory infiltrate and Civatte bodies/apoptotic keratinocytes at the epidermal-dermal junction (black arrow in microscopic image). Which of the following is true?

A. Junctional melanocyte proliferation is a prominent and constant feature
B. The primary inflammatory pattern seen in the microscopic images is spongiotic dermatitis
C. This disease is restricted to cutaneous sites and never involves the oral mucosa
D. This lesion is associated with hepatitis virus C infection and chronic active hepatitis
E. This patient has pustular psoriasis





The correct answer is “this lesion is associated with hepatitis virus C infection and chronic active hepatitis.” This is a case of lichen planus, which is associated with hepatitis C and chronic active hepatitis in about 30% of the cases. The primary inflammatory pattern seen in the microscopic images is known as lichenoid dermatitis (with a band-like lymphocytic infiltrate that obscures the dermoepidermal junction and is associated with epidermal basal cell damage; these apoptotic basal epidermal cells are known as “Civatte bodies” (see the black arrow in the 1st microscopic image) which are brightly eosinophilic round anucleate structures. There is not significant features to suggest that this is spongiotic dermatitis (would show more prominent intercellular edema). The 2nd microscopic image also shows a characteristic “saw-toothing” of the rete ridges, partially as a result of the inflammatory infiltrate damaging the basal layer. Oral involvement is common in lichen planus, affecting up to 60% of patients.

12. What is the most common malignancy in White populations?
    A. Sebaceous carcinoma
    B. Lung adenocarcinoma
    C. Basal cell carcinoma
    D. Melanoma
    E. Keratoacanthoma

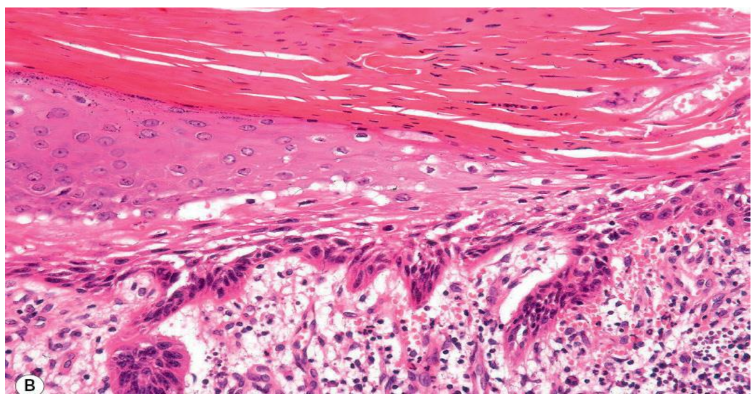
The correct answer is “basal cell carcinoma.”

13. A 75-year-old woman who has been a gardener her entire life has multiple scaly papules on the head, neck, and extremities (see clinical image, which shows a representative lesion on her dorsal hand). One of the lesions is biopsied, and a representative area of the corresponding H&E-stained section is shown in the

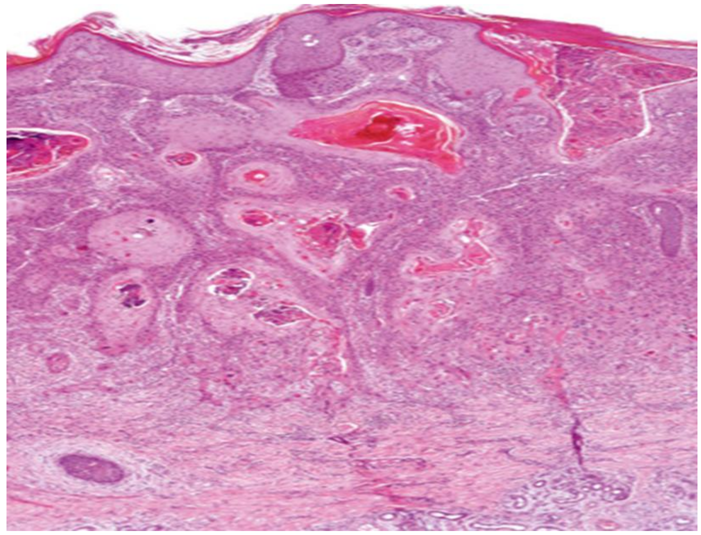
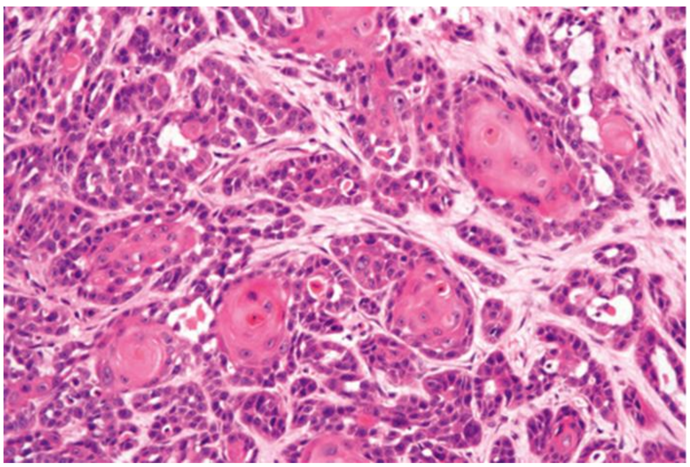
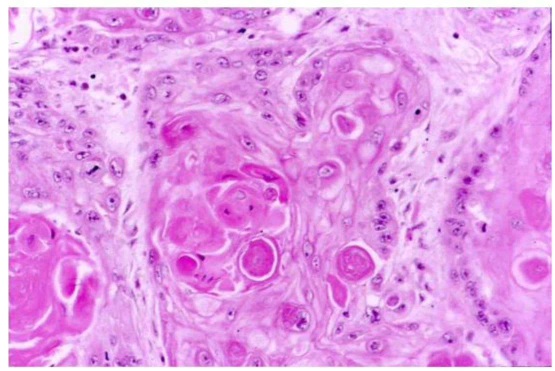
microscopic image below. Which of the following describes the features of this lesion the best?

A. Most patients with lesions of this type have an inherited genetic syndrome of defective nucleotide excision repair
B. Progression of this lesion into invasive carcinoma occurs at a rate of approximately 0.1 – 2.6% per year
C. The most common predisposing factor for the development of lesions of this type is exposure to arsenicals
D. This lesion is a keratoacanthoma and often regresses spontaneously
E. This lesion is a direct precursor to basal cell carcinoma

The correct answer is "progression of this lesion into invasive carcinoma occurs at a rate of approximately 0.1 – 2.6% per year." This is an actinic keratosis, which is characterized by squamous dysplasia that does not involve the full thickness of the epidermis.



14. An 85-year-old woman with venous insufficiency and peripheral arterial disease has had recalcitrant stasis dermatitis complicated by ulceration for more than 30 years. You have been astute about taking clinical photographs during her prior visits and notice that, in comparison, the ulcer now shows everted and indurated wound margins and bleeds to the touch (see clinical image). You decide to perform a biopsy within this everted and indurated area and examine the H&E-stained sections (see microscopic images). What is the risk factor that most likely contributed to the development of the lesion shown in the microscopic images?
A. Molluscum contagiosum causing immunosuppression
B. Chronic ulceration leading to chronic inflammation
C. Psoriasis causing severe pruritus
D. BRAF mutation causing clonal expansion of nevus cells
E. Recent diagnosis of mycosis fungoides that is limited in distribution

The correct answer is “chronic ulceration leading to chronic inflammation. “The clinical history, presentation, and histologic features are classic for “Marjolin’s ulcer” which is a type of cutaneous squamous cell carcinoma that arises in sites of chronic wounds; chronically inflamed skin (and therefore chronic wounds) are risk factors for developing cutaneous squamous cell carcinoma. The question stem describes a 30+ year course of chronic ulceration, which is the risk factor most likely contributing to the development of the squamous cell carcinoma in this case as opposed to a recent diagnosis of mycosis fungoides which commonly exhibits an indolent clinical course. As such, a recent diagnosis of mycosis fungoides that is limited in distribution would portend an excellent prognosis and would not majorly contribute to the development of squamous cell carcinoma in this setting.

15. A 40-year-old female presents with scattered brown colored papules on the lower extremities which she mentions arose in the setting of insect bites. When the lesion is

pinched, there is a central “dimpling” seen. One of the lesions is biopsied, and a representative image of the corresponding H&E-stained section demonstrated a bland dermal proliferation of spindle cells with collagen trapping. Which of the following is the most likely diagnosis?

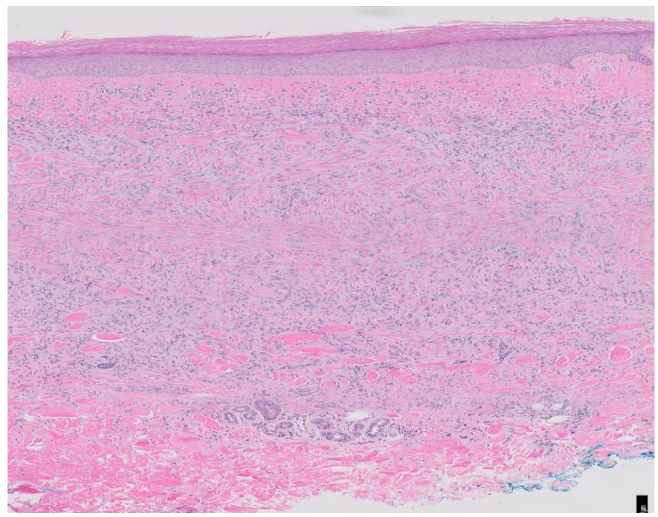
A. Dermatofibrosarcoma protuberans

B. Intradermal nevus

C. Benign dermatofibroma (fibrous histiocytoma)

D. Fibroepithelial polyp (skin tag)

E. Keloid

The correct answer is “benign dermatofibroma.” The clinical history, in which the nodules occurred in the setting of insect bites in conjunction with the positive “dimple” sign when pinched, supports the diagnosis. The microscopic image shows dermatofibroma, which is a dermal proliferation of bland spindle cells with evidence of “collagen trapping” towards the base of the lesion. The lesion is described as a papule, as opposed to an indurated plaque with irregular borders which would have raised the possibility of a dermatofibrosarcoma protuberans. An intradermal nevus would be comprised of nests of bland melanocytes within the dermis as opposed to a spindle cell proliferation. Fibroepithelial polyp would present as a pedunculated skin tag and the pathology would likely appear similar to normal skin. Keloid would appear as a scar that grows beyond the confines of the original wound as opposed to a well-delineated papule seen in the image for this case.

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References, including for the clinical images, may be found in the pertinent sections of the Dermatopathology powerpoint presentation.